# Latitude and Doubt

An exact ledger of meaning in messages

with measurements on the genetic code and on English

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*"Expression is free exactly where meaning is silent.*
*What the speaker chooses there is the speaker."*
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**Abstract.** Shannon set the meaning of messages aside as irrelevant to the engineering problem. This paper restores it as bookkeeping. For a source that pairs what is meant, $W$, with what is said, $X$, the joint entropy divides exactly into three non-negative parts: the information the form conveys, $I(W;X)$; the *latitude* $\Lambda = H(X \mid W)$, which the speaker remains free to choose once the meaning is fixed; and the *doubt* $\Delta = H(W \mid X)$, which the listener cannot recover from the form. The identities are the chain rule; the use made of them is the contribution. Meaning need not be declared by an annotator. For any purpose $Y$, the coarsest partition of messages that loses no information about $Y$ is the belief $P(Y \mid x)$ that they induce, and only the identity partition is adequate for every purpose. The *leakage* $I(X;Y \mid W)$ measures how much a declared meaning fails to screen off form from consequence. Latitude has an operational value: in an idealized model with a shared key, it is the exact rate at which a hidden message can be written into the form of an unambiguous source without altering the meaning or the statistics of the text, and a finite-precision implementation is analysed separately. An unambiguous language with content $C$ and latitude $\Lambda$ needs at least $2^{C+\Lambda}$ distinct forms. We measure the ledger on six sequenced genomes, where the meaning map is the genetic code and the doubt is zero. Latitude is 1.235–1.501 bits per codon; synonymous choice carries 88.5 % of the information that identifies the genome of a codon and 90.2 % of the information that separates highly expressed *E. coli* genes from the rest; held-out genes reproduce these information values to within 0.003 bits; and 90,127 bits were hidden in 240 real genes without changing the encoded amino-acid sequence of any of them. In English the balance between latitude and doubt depends on what counts as a form and on whether items are weighted by use: on the well-attested core of SemCor doubt exceeds latitude for lemmas, by 0.44 bits per word, but not for inflected words, and among 22 wordnets latitude exceeds doubt in 12. The estimators are validated against a law of known truth, and every number in the text is produced by the code that accompanies the paper.

**Keywords.** semantic information, entropy, synonymy, ambiguity, sufficiency, leakage, genetic code, steganography.
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## 1 Introduction

Shannon opened his theory of communication by taking meaning out of it: "These semantic aspects of communication are irrelevant to the engineering problem" [51]. The sentence describes a problem, not the world. Weaver, introducing the theory to a wider readership, separated a technical problem (how accurately symbols are transmitted), a semantic problem (how precisely they convey the intended meaning) and an effectiveness problem (how well the received meaning affects conduct) [56]. The first has a complete solution. The second has drawn many proposals [3, 4, 23, 35], but none with the plain, exact accounting that Shannon gave the first.

This paper gives such an accounting. Take a source in which every message has a form $X$, what is said, and a meaning $W$, what is meant. Two facts hold of every language. Many forms can carry one meaning: synonymy. One form can carry many meanings: ambiguity. The entropy $H(X)$ of the forms mixes the two consequences. It counts the meaningful choices the speaker makes together with the meaningless freedom the speaker keeps, and it is silent about what the listener cannot recover. We separate them. Call $C = H(W)$ the *content*, $E = H(X)$ the *expression*, $\Lambda = H(X \mid W)$ the *latitude*, $\Delta = H(W \mid X)$ the *doubt* and $I = I(W;X)$ the *conveyance*. Then

$$E = I + \Lambda, \qquad C = I + \Delta, \qquad \text{hence} \qquad E - C = \Lambda - \Delta.$$

What is said exceeds what is meant by exactly the freedom of the speaker minus the doubt of the listener (Theorem 1).

The ledger is the chain rule of entropy applied to a pair, and no more. The paper asks what it is good for. The answers, in the order they appear, are these.

1. *Meaning without an annotator* (Section 3). If $Y$ is what a receiver will act on, the meaning of a message for that purpose is the belief $P(Y \mid x)$ it induces. This partition is the coarsest that loses no information about $Y$ (Theorem 4), it shrinks as the purpose weakens, and no partition is adequate for every purpose. The *leakage* $I(X;Y \mid W)$ tests a declared meaning against a purpose (Theorem 5).
2. *What the terms cost and buy* (Section 4). Sending the form and then resolving the meaning costs $\Lambda$ bits more than sending the meaning, whatever the doubt (Theorem 7). In an idealized model with a shared key, latitude is a channel: exactly the number of bits per symbol that can be written into the form without changing the meaning or the statistics of the text (Theorem 8); the finite-precision implementation is analysed separately (Proposition 9). An unambiguous language with latitude $\Lambda$ needs $2^{C+\Lambda}$ forms (Corollary 2).
3. *Estimation* (Section 5). The plug-in ledger keeps every identity; its terms are biased, and we quantify how, with one certified interval and one validated protocol.
4. *Measurement* (Section 6). The genetic code is the one natural language whose meaning map is physically implemented and exactly known. On six genomes we measure the ledger, the leakage about two purposes, and the capacity of latitude on real genes. In English, where synonymy and ambiguity both occur, we measure the ledger on a sense-tagged corpus and on sense inventories, and test how the answer depends on the definition of the form.

**What is and is not new.** The identities of Theorem 1 are the chain rule, and for a deterministic meaning map they are the grouping property that Shannon stated in the

**Table 1.** Results and what each rests on.

| Result | Statement | Rests on |
|---|---|---|
| Theorem 1 | $E = I + \Lambda$, $C = I + \Delta$, $E - C = \Lambda - \Delta$ | chain rule |
| Corollary 2 | an unambiguous language needs at least $2^{C+\Lambda}$ distinct forms | $H(X) \leq \log N$ |
| Proposition 3 | $\Lambda + \Delta$ is a distance between meaning and form | conditioning reduces entropy |
| Theorem 4 | meaning for purpose $Y$ is the belief $P(Y \mid x)$: sufficient, coarsest, monotone in $Y$ | sufficiency; chain rule |
| Theorem 5 | leakage $I(X;Y \mid W)$; abstraction leaks the relevance it discards; only injective maps are adequate for every purpose | chain rule; data processing |
| Proposition 6 | entropy and relevance each split over the levels of meaning | chain rule |
| Theorem 7 | saying instead of meaning costs $\Lambda$ bits per symbol, whatever the doubt | source coding theorem |
| Theorem 8, Proposition 9 | latitude is a channel of capacity $\Lambda$ bits per symbol in an idealized model, with exact failure probability; bounds for a finite-precision implementation | Fano's inequality; arithmetic coding |
| Proposition 10 | certified interval for the entropy of a small alphabet | $L_1$ concentration; continuity of entropy |
| Section 6 | six genomes, held-out genes, embedding in real genes, SemCor, verification | data and code supplied |

same paper [51]. The canonical meaning of Theorem 4 is the minimal sufficient statistic [7, 22] in the vocabulary of the information bottleneck [55] and of causal states [50]. The price of saying is Shannon's source coding theorem, and latitude as a channel is the capacity of provably secure steganography [10, 30] written in the ledger's terms. What we offer is the vocabulary in which these facts read as one account, the purpose-relative definition of meaning with the leakage and depth identities that follow from it, the vocabulary law, an exact failure probability for the latitude channel, and measurements on real data. Table 1 lists each result with what it rests on, so that no statement is mistaken for more than it is.

**Reproducibility.** Every table entry and every number quoted in the text is written by scripts in the supplied archive from the raw sequence and corpus files (Section A); none was transcribed.

## 2 The ledger

### 2.1 Setting and definitions

Alphabets are finite, logarithms are to base 2 and entropies are in bits. $H$ and $I$ denote Shannon entropy and mutual information [14]; $\mathbb{P}$ is probability.

**Definition 1** (Semantic source)**.** A *semantic source* is a pair of random variables $(W, X)$ with finite ranges $\mathcal{W}$ and $\mathcal{X}$ and a joint law $P$. $W$ is the *meaning*, what is meant; $X$ is the *form*, what is said. Both ranges are restricted to the support of their marginals.

No function between $\mathcal{X}$ and $\mathcal{W}$ is assumed. The joint law is the language: $P(x \mid w)$ says how meanings are said, and $P(w \mid x)$ says how forms are understood. Synonymy is a meaning with more than one likely form, ambiguity a form with more than one likely meaning; both may be present.

**Definition 2** (Ledger)**.** The *ledger* of a semantic source has five entries.

| Name | Symbol | Definition | Reads as |
|---|---|---|---|
| Content | $C$ | $H(W)$ | what is meant |
| Expression | $E$ | $H(X)$ | what is said |
| Latitude | $\Lambda$ | $H(X \mid W)$ | freedom of the speaker once the meaning is fixed |
| Doubt | $\Delta$ | $H(W \mid X)$ | what the listener cannot recover from the form |
| Conveyance | $I$ | $I(W;X)$ | what the form does convey |

Each entropy has a multiplicative reading in perplexity, the effective number of equally likely outcomes [29, 32]: $2^{\Lambda}$ is the effective number of ways a meaning is said, $2^{\Delta}$ the effective number of meanings a form is taken to have.

**Theorem 1** (The ledger)**.** *For every semantic source:*

(a) $E = I + \Lambda,\ C = I + \Delta,\ H(W,X) = I + \Lambda + \Delta$;

(b) $E - C = \Lambda - \Delta$;

(c) $\Lambda \geq 0$, $\Delta \geq 0$, $0 \leq I \leq \min\{C, E\}$;

(d) $\Delta = 0$ *if and only if* $W = \pi(X)$ *almost surely for a function* $\pi$; $\Lambda = 0$ *if and only if* $X = \xi(W)$ *almost surely for a function* $\xi$; $I = 0$ *if and only if* $W$ *and* $X$ *are independent;*

(e) *if* $\Delta = 0$, *then* $E = C + \Lambda$ *with* $C = H(\pi(X))$ *and* $\Lambda = \sum_w \mathbb{P}(w)\, H(X \mid W{=}w)$.

*Proof.* (a) The chain rule gives $H(W,X) = H(W) + H(X \mid W) = H(X) + H(W \mid X)$, and $I = H(W) - H(W \mid X) = H(X) - H(X \mid W)$. Hence $E = I + \Lambda$, $C = I + \Delta$, and $H(W,X) = C + \Lambda = I + \Delta + \Lambda$. (b) Subtract the first two identities of (a). (c) Conditional entropy and mutual information are non-negative; $I \leq C$ because $\Delta \geq 0$ and $I \leq E$ because $\Lambda \geq 0$. (d) $H(W \mid X) = \sum_x \mathbb{P}(x) H(W \mid X{=}x)$ vanishes iff every term does, iff each conditional law of $W$ is a point mass, iff $W$ is a function of $X$; the case of $\Lambda$ is symmetric; $I = 0$ is independence. (e) Set $\Delta = 0$ in (a); the formulas follow from the definitions. □

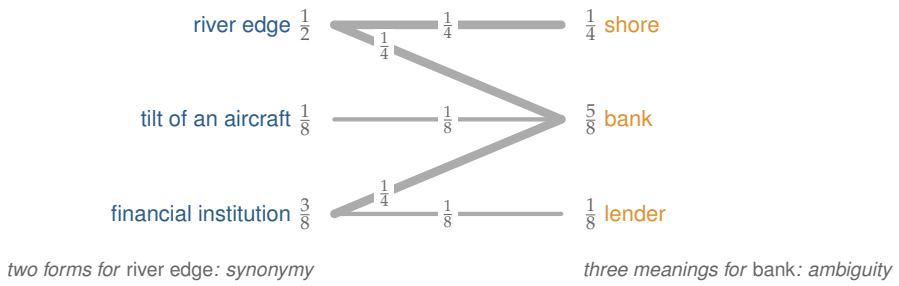
Parts (a)–(c) are the information diagram of two variables [63]; part (e) is Shannon's grouping property [51]. They are stated here because every later result is a reading of them.

**Corollary 2** (Vocabulary law)**.** *If* $X$ *takes at most* $N$ *values, then* $E \leq \log N$, *with equality iff* $X$ *is uniform, and*

$$\Delta - \Lambda = C - E \ \geq\ C - \log N.$$

*In particular an unambiguous language* ($\Delta = 0$) *with content* $C$ *and latitude* $\Lambda$ *uses an effective vocabulary* $2^E = 2^{C+\Lambda} \leq N$: *each bit of freedom of expression doubles the number of forms the language needs. When the vocabulary is saturated* ($E = \log N$), *every additional bit of latitude is paid for by exactly one bit of doubt.*

**(a) A lexicon is a joint law of meaning and form; line width is probability**




**(b) The same law as a ledger; one bit is 3.4 cm**

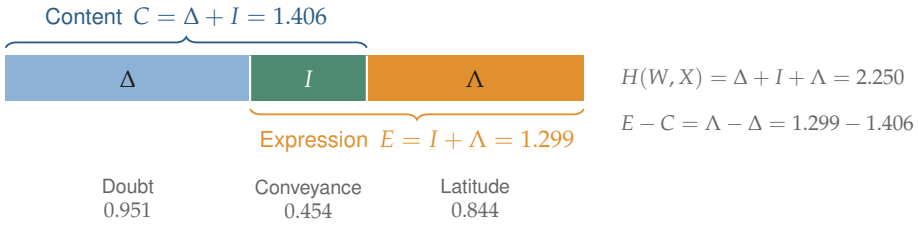



**Figure 1.** The ledger of a small lexicon. (a) Three meanings and three forms; the form *bank* carries three meanings, the meaning *river edge* two forms. (b) The joint entropy is cut into doubt $\Delta = H(W \mid X)$, conveyance $I = I(W;X)$ and latitude $\Lambda = H(X \mid W)$. Content is the first two parts, expression the last two; the overlap is what the form conveys. All quantities in bits.

*Proof.* $H(X) \le \log N$ with equality iff $X$ is uniform is the maximum-entropy property. The rest is Theorem 1(b) and (e). □

**Proposition 3** (Distance between meaning and form)**.** $d(W,X) = \Lambda + \Delta = H(W,X) - I(W;X)$ *is symmetric, non-negative, vanishes iff* $W$ *and* $X$ *are functions of each other, and satisfies* $d(A,C) \le d(A,B) + d(B,C)$ *for any three random variables on one probability space.*

*Proof.* $H(A \mid C) \le H(A,B \mid C) = H(B \mid C) + H(A \mid B,C) \le H(B \mid C) + H(A \mid B)$, and likewise with $A$ and $C$ exchanged; add the two inequalities. □

The quantity $H(A \mid B) + H(B \mid A)$ is Rokhlin's distance between partitions [48] and the variation of information used to compare clusterings [38]. Here it measures how far a language is from a one-to-one correspondence between meanings and forms.

## 2.2 A lexicon

Figure 1 shows a lexicon with three meanings and three forms. The form *bank* carries three meanings, the meaning *river edge* two forms. The joint entropy is 2.250 bits. Content is $C = 1.406$ bits and expression $E = 1.299$ bits, so more is meant than is said. The balance $E - C = \Lambda - \Delta = 0.844 - 0.951$ is negative because doubt (0.951 bits) exceeds latitude (0.844 bits). The form conveys $I = 0.454$ bits of the meaning. Nothing in the numbers depends on the words; a different assignment of the same probabilities gives the same ledger.

### 2.3 The genetic code

The genetic code is the cleanest natural example, because its meaning map is a physical mechanism. A codon $X$ is translated by the ribosome into an amino acid or a stop signal $W$; the first assignment was deciphered by Nirenberg and Matthaei [43], and the pairing rules behind its many-to-one structure were described by the wobble hypothesis [15]. Each codon has one translation, so $\Delta = 0$ and Theorem 1(e) applies: the entropy of the codon is the entropy of the protein plus the entropy of the synonymous choice. For the *E. coli* K-12 genome (Section 6) $C = 4.175$ bits, $\Lambda = 1.501$ bits and $E = 5.676$ bits per codon. The effective vocabulary is $2^E = 51.1$ of the 64 codons, in agreement with Corollary 2: a language that must be unambiguous and has this much freedom cannot use fewer forms. Sending the protein instead of the DNA saves 26.5 % of the entropy.

## 3 Where meaning comes from

A ledger needs a meaning $W$. There are two ways to obtain one. Someone can declare it, or it can be derived from what the message is for.

### 3.1 Declared meaning

In many systems $W$ is attached to $X$ by a process outside the source. The genetic code is a physical process. A lexicographer's sense inventory is a human one: WordNet assigns each word occurrence a synset [19, 40]. The behaviour of a program on a test suite is an operational one. In each case the ledger of Section 2 applies as it stands. What it cannot say is whether the declared $W$ is the right one for a given use. That is the question of the next two subsections.

### 3.2 Meaning for a purpose

Wittgenstein wrote that the meaning of a word is its use in the language [59], and Bateson called information a difference which makes a difference [5]. Both point to consequences. We take them literally.

**Definition 3** (Purpose, belief, meaning for a purpose)**.** Let $(X, Y)$ be jointly distributed with finite ranges; $Y$ is the *purpose*, whatever the receiver will act on or learn. For $x$ in the support of $X$ the *belief* is $b(x) = \mathbb{P}(Y \in \cdot \mid X{=}x)$, a distribution on $\mathcal{Y}$. The *meaning of $X$ for purpose $Y$* is the random element $M_Y = b(X)$; equivalently, the partition of the support of $X$ into the level sets of $b$.

Two messages have the same meaning for $Y$ exactly when they leave the receiver with the same belief about $Y$. This depends only on $P_{XY}$; no annotator is involved.

**Theorem 4** (Meaning for a purpose)**.** *Let $M = M_Y$.*

(a) Sufficiency. *$Y - M - X$ is a Markov chain. Hence $I(X;Y \mid M) = 0$ and $I(M;Y) = I(X;Y)$.*

(b) Minimality. *If $T = f(X)$ is any function with $I(T;Y) = I(X;Y)$, then $M$ is a function of $T$ almost surely. So $M$ is the coarsest function of $X$ that loses no information about $Y$; it is unique up to relabelling.*

(c) Ledger. *For the pair $(M, X)$ the doubt is zero and*

$$E = H(M) + H(X \mid M) = \underbrace{I(X;Y)}_{\textit{consequence}} + \underbrace{H(M \mid Y)}_{\textit{hedge}} + \underbrace{H(X \mid M)}_{\textit{latitude}}.$$

(d) Monotone in the purpose. *If $Y' = h(Y)$ for a function $h$, then $M_{Y'}$ is a function of $M_Y$; so $H(M_{Y'}) \le H(M_Y)$ and $H(X \mid M_{Y'}) \ge H(X \mid M_Y)$.*

(e) Extremes. *If $Y$ is constant, $M$ is constant and all of $E$ is latitude. If $Y = X$, then $M$ is in one-to-one correspondence with $X$ and the latitude is zero.*

*Proof.* (a) Let $m$ be a value of $M$ and $x$ any message with $b(x) = m$. Then $\mathbb{P}(y \mid x, m) = \mathbb{P}(y \mid x) = m(y)$, and $\mathbb{P}(y \mid m) = \sum_{x:\, b(x)=m} \mathbb{P}(x \mid m)\, b(x)(y) = m(y)$. So $\mathbb{P}(y \mid x, m) = \mathbb{P}(y \mid m)$, that is $Y \perp X \mid M$. Since $M$ is a function of $X$, the chain rule gives $I(X;Y) = I(X, M; Y) = I(M;Y) + I(X;Y \mid M) = I(M;Y)$. (b) Let $T = f(X)$ with $I(T;Y) = I(X;Y)$. Because $T$ is a function of $X$, $I(X;Y \mid T) = I(X, T;Y) - I(T;Y) = I(X;Y) - I(T;Y) = 0$, so $Y \perp X \mid T$. Hence for every $t$ with positive probability and all $x, x'$ in $f^{-1}(t)$ of positive probability, $\mathbb{P}(y \mid x) = \mathbb{P}(y \mid t) = \mathbb{P}(y \mid x')$. Thus $b$ is constant on each fibre of $f$, that is $M = g(T)$. (c) $M$ is a function of $X$, so $H(M \mid X) = 0$; the chain rule gives $H(X) = H(X, M) = H(M) + H(X \mid M)$. By (a), $H(M) = I(M;Y) + H(M \mid Y) = I(X;Y) + H(M \mid Y)$. (d) The belief about $Y'$ is the image of the belief about $Y$ under $h$, so $b(x) = b(x')$ implies $b'(x) = b'(x')$. A function of a variable has no more entropy, and conditioning on more leaves no more. (e) If $Y$ is constant, all beliefs coincide. If $Y = X$, then $b(x)$ is the point mass at $x$, which determines $x$. □

The consequence term $I(X;Y)$ is the information about $Y$ that the message carries, and it is the only one of the three that grows when the message becomes more informative. The hedge $H(M \mid Y)$ is the entropy of the receiver's confidence not explained by the truth: a forecaster who says "30 %" on some days and "70 %" on others has content 1 bit in $M$ but conveys less than one bit about rain. Content, in this sense, counts distinct beliefs; it is not the same as information. The latitude is what remains: variation of the form to which the purpose is blind.

### 3.3 Adequacy and leakage

A declared meaning $W$ is adequate for a purpose when it screens the form off from the consequence.

**Definition 4** (Adequacy and leakage)**.** Let $W$ be jointly distributed with $(X, Y)$. The *leakage* of $W$ about the purpose $Y$ is $\mathcal{L}_Y(W) = I(X;Y \mid W)$. The meaning $W$ is *adequate* for $Y$ if $\mathcal{L}_Y(W) = 0$.

**Theorem 5** (Leakage)**.** (a) $\mathcal{L}_Y(W) \ge 0$, *with equality iff $Y \perp X \mid W$.*

(b) *If $W = f(X)$, then $\mathcal{L}_Y(W) = I(X;Y) - I(W;Y)$, and $W$ is adequate for $Y$ iff $M_Y$ is a function of $W$, that is, iff the partition induced by $f$ refines the partition induced by $b$. In general $\mathcal{L}_Y(W) = I(X;Y) - I(W;Y) + I(W;Y \mid X)$.*

(c) Abstraction leaks. *If $W_1 = f(X)$ and $W_2 = g(W_1)$, then $\mathcal{L}_Y(W_2) = \mathcal{L}_Y(W_1) + I(W_1;Y \mid W_2)$: each step of abstraction leaks exactly the relevance it discards.*

(d) No universal meaning. *If $W = f(X)$ is adequate for every purpose $Y$, then $f$ is injective on the support of $X$.*

*Proof.* (a) is the non-negativity of conditional mutual information. (b) For $W = f(X)$ the chain rule gives $I(X;Y) = I(X, W;Y) = I(W;Y) + I(X;Y \mid W)$. In general $I(X, W;Y)$ expands in both orders,

$$I(W;Y) + I(X;Y \mid W) = I(X;Y) + I(W;Y \mid X),$$

which is the last formula. If $I(W;Y) = I(X;Y)$ then Theorem 4(b) gives $M = g(W)$. Conversely, if $M = g(W)$ then $I(W;Y) \geq I(M;Y) = I(X;Y)$ by data processing, and $I(W;Y) \leq I(X;Y)$ because $W$ is a function of $X$. (c) By (b), $\mathcal{L}_Y(W_2) - \mathcal{L}_Y(W_1) = I(W_1;Y) - I(W_2;Y)$, and $I(W_1;Y) = I(W_1,W_2;Y) = I(W_2;Y) + I(W_1;Y \mid W_2)$ since $W_2$ is a function of $W_1$. (d) Take $Y = X$. Then $\mathcal{L}_X(W) = I(X;X \mid W) = H(X \mid W)$, which vanishes iff $X$ is a function of $W = f(X)$, iff $f$ is injective on the support. □

Part (d) is the philosophical content of the section. There is no purpose-free meaning: a partition that discards anything discards it for some purpose. The trivial partition is adequate for the purpose "nothing", and the identity is adequate for the purpose "the exact words". Every meaning in between is a statement about what the receiver cares about.

The leakage has an everyday reading. Linguists call the part of a word's effect that its denotation does not fix its connotation, register or style. Leakage measures it in bits, for a stated purpose: how much a hearer learns about the consequence from the choice among synonyms.

### 3.4 Depth

Meanings come in levels. A codon has an amino acid, which has a chemical class; a word occurrence has a sense, which has a semantic field.

**Proposition 6** (Depth ledger)**.** *Let $W_1 = f_1(X)$, $W_2 = f_2(W_1)$, …, $W_L = f_L(W_{L-1})$ be successive functions, and $W_0 = X$. Then*

$$H(X) = H(W_L) + \sum_{k=1}^{L} H(W_{k-1} \mid W_k), \qquad I(X;Y) = I(W_L;Y) + \sum_{k=1}^{L} I(W_{k-1};Y \mid W_k).$$

*The term with $k = 1$, the first of each sum, is the latitude $H(X \mid W_1)$ and the leakage $\mathcal{L}_Y(W_1)$, respectively: the contribution of the finest level.*

*Proof.* Since $W_k$ is a function of $W_{k-1}$, the chain rule gives

$$H(W_{k-1}) = H(W_k) + H(W_{k-1} \mid W_k), \qquad I(W_{k-1};Y) = I(W_k;Y) + I(W_{k-1};Y \mid W_k).$$

Sum over $k$ and telescope. □

Figure 2 shows the two ledgers side by side. They are two decompositions of two different quantities, the entropy of the message and the information it carries about a purpose, over the same levels. The first says where the choices are made; the second says where the consequences live. In Section 6 the two profiles differ sharply: the entropy of a codon is spread over all three levels, while the information it carries about which genome it came from sits almost entirely at the finest level, the synonymous choice.

## 4 What the terms cost and buy

### 4.1 The price of saying

Let $(W_i, X_i)$, $i = 1, \ldots, n$, be independent copies of a semantic source. A speaker who holds $W^n$ can send it directly, or can send the forms $X^n$ and then whatever the listener still needs to fix $W^n$. Both are lossless descriptions of the meaning. Their costs differ by exactly the latitude.

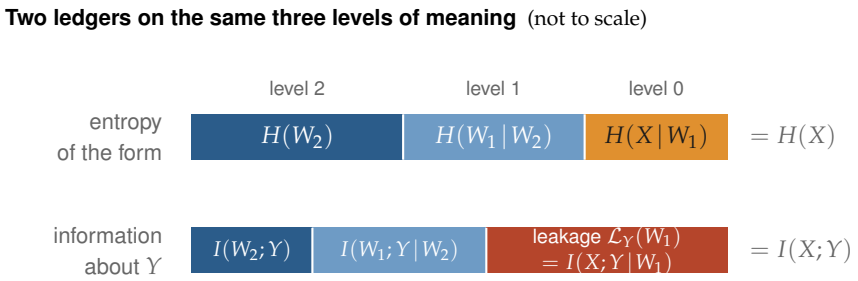



**Figure 2.** Depth. If $W_1 = f(X)$ and $W_2 = g(W_1)$ are successively coarser meanings, the entropy of the form is a sum with one term per level (Proposition 6); the information the form carries about a purpose $Y$ splits over the same levels, and its finest-level term, drawn at the right, is the leakage of $W_1$ (the $k = 1$ term of the sum). The two bars have different scales.

**Theorem 7** (The price of saying)**.** *Let $\ell_W$, $\ell_X$ and $\ell_{W|X}$ be the smallest expected lengths, in bits, of a prefix code for $W^n$, of a prefix code for $X^n$, and of a prefix code for $W^n$ that may depend on $x^n$. Then*

$$nC \le \ell_W < nC + 1, \qquad nE \le \ell_X < nE + 1, \qquad n\Delta \le \ell_{W|X} < n\Delta + 1.$$

*Sending $X^n$ and then resolving the meaning therefore costs between $n(E+\Delta)$ and $n(E+\Delta)+2$ bits, which is*

$$n(E + \Delta) = nC + n\Lambda.$$

*Saying instead of meaning costs $n\Lambda$ bits, up to two, whatever the doubt.*

*Proof.* For a random variable $Z$ the expected length of an optimal prefix code lies in $[H(Z), H(Z)+1)$ [14, 31]. Apply this to $Z = W^n$, to $Z = X^n$, and, for each $x^n$, to the conditional law of $W^n$ given $X^n = x^n$; averaging over $x^n$ gives a length in $[H(W^n \mid X^n), H(W^n \mid X^n)+1)$. Independence gives $H(W^n) = nC$, $H(X^n) = nE$ and $H(W^n \mid X^n) = n\Delta$. The identity $E + \Delta = C + \Lambda$ is Theorem 1. □

The doubt has its own operational reading. A listener who has heard $x$ and asks yes/no questions of an oracle to fix the meaning needs, by an optimal question tree, between $H(W \mid X{=}x)$ and $H(W \mid X{=}x) + 1$ questions on average; averaged over forms this is between $\Delta$ and $\Delta + 1$. Doubt is the price of clarification, and latitude is the price of wording. When $\Delta = 0$, as in the genetic code, saying instead of meaning costs $n\Lambda$ and nothing is gained by asking.

### 4.2 Latitude is a channel

Latitude is what the speaker may choose without changing the meaning. It can carry a second message. The question is how much, if the text must still mean the same thing and must still look like text. We answer first in an idealized model, with its assumptions stated, and then treat the finite-precision implementation separately (Proposition 9).

**The idealized model.** Let $(W_i, X_i)$ be independent copies of a source with $\Delta = 0$, so $W = \pi(X)$ and $Q(x^n \mid w^n) = \prod_i P(x_i \mid w_i)$ is the *cover law*. A *k-bit embedding scheme* has a key $R$ shared with the receiver, a hidden message $B$ uniform on $\{0,1\}^k$, and private randomness of the sender. The encoder maps $(w^n, B, R, \text{randomness})$ to a form sequence $\tilde{X}^n$ such that (i) $\pi^n(\tilde{X}^n) = w^n$ (the meaning is preserved) and (ii) for every $w^n$ and every message the law of $\tilde{X}^n$ is exactly $Q(\cdot \mid w^n)$ (the text is indistinguishable from the cover

law). The decoder outputs $\hat{B}$ from $(\tilde{X}^n, R)$, or reports failure; failure counts as an error below, with probability $\delta$. The assumptions are these.

(A1) The cover law $Q$ is known exactly to the sender and to the receiver.

(A2) The meaning sequence $w^n$ is fixed by the source, not by the hidden message, and the receiver computes it from the text; this is where $\Delta = 0$ is used.

(A3) The key $R$ is uniform on $[0, 1)$, secret, used once, and independent of the message, the meaning sequence and the sender's randomness.

(A4) Real numbers and exact arithmetic are available.

Security here means (ii): against an observer who knows $Q$ and does not know $R$, the text has exactly the law of an unmarked text. It is relative to $Q$; an observer with a better model of natural text is outside the guarantee. If the message is itself uniformly random, for instance a ciphertext, (ii) holds without a secret key, and $R$ only randomizes the alignment on which the failure probability below is computed; it may then be public, and the failure probability is an average over $R$.

**Theorem 8** (Latitude is a channel)**.** *Under (A1)–(A4):* Converse. *Every scheme satisfying (i) and (ii) obeys* $(1-\delta)k \le n\Lambda + 1$ *when the meanings* $W_i$ *are drawn from the source, and* $(1-\delta)k \le \sum_i H(X \mid W{=}w_i) + 1$ *for a fixed meaning sequence* $w^n$*.* Achievability. *For a fixed* $w^n$ *put* $\mu = \sum_i H(X \mid w_i)$ *and* $\sigma^2 = \sum_i \mathrm{Var}[-\log P(X \mid w_i)]$*. For every integer* $k \ge 0$ *the scheme given below satisfies (i) and (ii); its decoder either outputs the hidden message or reports failure, and never outputs a wrong message; and its failure probability is*

$$\delta = \sum_{x^n} Q(x^n \mid w^n)\ \min\{1,\, 2^k Q(x^n \mid w^n)\},$$
$$\delta \le \mathbb{P}\big[-\log Q(X^n \mid w^n) \le k + s\big] + 2^{-s} \qquad \text{for every } s \ge 0.$$

*With* $k = \lfloor \mu - a \rfloor - s$ *Chebyshev's inequality gives* $\delta \le \sigma^2/a^2 + 2^{-s}$*; taking* $a = n^{3/4}$ *and* $s = \lceil \log n \rceil$*, the rate* $k/n$ *tends to* $\Lambda$ *and* $\delta \to 0$*. In words:* the capacity of the free channel inside a message is its latitude.

*Proof of the converse.* $B$ is independent of $(W^n, R)$, so $k = H(B \mid W^n)$. Write $k = I(B; \tilde{X}^n, R \mid W^n) + H(B \mid \tilde{X}^n, R, W^n)$. Since $\hat{B}$ is a function of $(\tilde{X}^n, R)$, Fano's inequality gives $H(B \mid \tilde{X}^n, R, W^n) \le 1 + \delta k$ ($W^n$ is a function of $\tilde{X}^n$). Next $I(B; \tilde{X}^n, R \mid W^n) = I(B; R \mid W^n) + I(B; \tilde{X}^n \mid R, W^n) = 0 + I(B; \tilde{X}^n \mid R, W^n) \le H(\tilde{X}^n \mid W^n)$. By (ii), $H(\tilde{X}^n \mid W^n{=}w^n) = \sum_i H(X \mid w_i)$; averaging over $W^n$ gives $n\Lambda$. Combine. □

*Proof of achievability. Scheme.* Fix a total order on $\mathcal{X}^n$ and let $I(x^n) = [F(x^n), F(x^n) + Q(x^n \mid w^n)) \subset [0, 1)$ be its interval in the cumulative distribution of the cover law. Given the message $m \in \{0, \ldots, 2^k - 1\}$, draw $V$ uniform on $[0, 1)$ and put $z = (m + V)2^{-k}$ and $U = (z + R) \bmod 1$. The stego sequence $\tilde{X}^n$ is the $x^n$ with $U \in I(x^n)$. *Security.* $R$ is uniform and independent of $z$, so $U$ is uniform, whence $\mathbb{P}[\tilde{X}^n = x^n] = Q(x^n \mid w^n)$ for every $m$. Property (ii) holds exactly; (i) holds because $\tilde{X}^n$ is drawn from the cover law given $w^n$. *Decoding.* The receiver knows $w^n$ (it is $\pi^n(\tilde{X}^n)$), $R$, and hence the interval $I = I(\tilde{X}^n)$ of width $q = Q(\tilde{X}^n \mid w^n)$ that contains $U$. It forms the arc $A = \{(u - R) \bmod 1 : u \in I\}$, which contains $z = (U - R) \bmod 1$. If $A$ lies inside one cell $[j2^{-k}, (j+1)2^{-k})$ it outputs $j$, which then equals $m$ because $z$ lies in the cell of $m$; otherwise it reports failure. A wrong message is never output. *Failure probability.* Given $\tilde{X}^n$, $U$ is uniform on $I$; write $U = a + \rho q$ with $\rho$ uniform on $[0, 1)$ and independent of $V$. Then $A = [z - \rho q,\ z + (1 - \rho)q)$. In units of the cell width, $z$ sits at position $V$, so $A$ fits in the cell iff $\rho t \le V \le 1 - (1 - \rho)t$ with

$t = 2^k q$, which has probability $(1-t)^+$ for every $\rho$. Hence the failure probability given $\tilde{X}^n$ is $\min\{1, 2^k q\}$, and averaging over $\tilde{X}^n \sim Q$ gives the stated $\delta$. *Bound.* If $q \geq 2^{-k-s}$ bound $\min\{1, 2^k q\}$ by 1; otherwise by $2^k q < 2^{-s}$. This yields the inequality. For the Chebyshev step, $-\log Q(X^n \mid w^n)$ is a sum of independent terms with mean $\mu$ and variance $\sigma^2$, and $k + s \leq \mu - a$. Finally $\sigma^2 = O(n)$, so $\sigma^2/n^{3/2} \to 0$, and $\mu = n\Lambda + O(\sqrt{n})$ for typical meaning sequences, so $k/n \to \Lambda$. □

**Finite precision.** A computer has neither real numbers nor a perfectly uniform key. The next result separates the idealized theorem from an implementation with exact rational arithmetic and a key of $P$ bits.

**Proposition 9** (Finite precision)**.** *Replace $R$ and $V$ by independent variables uniform on the grid $G_P = \{j2^{-P} : 0 \leq j < 2^P\}$, keep all arithmetic exact, and let $q_{\min} = \min_{x^n} Q(x^n \mid w^n)$. For every message $m$:*

(a) *the decoder outputs $m$ or reports failure, as before;*

(b) $\left|\mathbb{P}[\tilde{X}^n = x^n] - Q(x^n \mid w^n)\right| \leq 2^{-P}$ *for every $x^n$; so if $P \geq \log(1/q_{\min}) + s + 1$, the total variation distance between the law of $\tilde{X}^n$ and $Q(\cdot \mid w^n)$ is at most $2^{-s-2}$;*

(c) *under the same condition, the failure probability $\delta_P$ differs from the idealized $\delta$ of Theorem 8 by at most $2^{-s} + 2^{k+1-P}$.*

*Proof.* (a) The decoding argument of Theorem 8 used only $U \in I$ and $z = (U - R) \bmod 1$, both exact here. (b) For fixed $(m, V)$, $U = (z + R) \bmod 1$ is uniform on the $2^P$ points of a translate of $G_P$. An interval of width $q$ contains $\lfloor q2^P \rfloor$ or $\lceil q2^P \rceil$ of them, so $\mathbb{P}[\tilde{X}^n = x^n \mid m, V]$ is within $2^{-P}$ of $q(x^n) = Q(x^n \mid w^n)$, and so is the average over $V$. If $P \geq \log(1/q_{\min}) + s + 1$ then $q(x^n) \geq 2^{s+1-P}$ for every $x^n$, so the relative error is at most $2^{-s-1}$ and the total variation distance is at most $\frac{1}{2} \cdot 2^{-s-1} = 2^{-s-2}$. (c) Couple the two schemes by drawing $(R', V')$ uniform on $[0,1)^2$ and setting $R = 2^{-P}\lfloor 2^P R' \rfloor$, $V = 2^{-P}\lfloor 2^P V' \rfloor$; the primed variables drive the idealized scheme. Then $U' - U \in [0, w)$ modulo 1 with $w = 2^{1-P}$. The two texts differ only if the left endpoint of an interval lies within $w$ below $U'$, which has probability $\sum_x \min\{q(x), w\} \leq w/q_{\min} \leq 2^{-s}$. If the texts agree, the two arcs differ by a shift $d < 2^{-P}$, and the outcomes differ only if a cell boundary falls in one of the two windows of length $d$ that follow the endpoints of the idealized arc. Each endpoint is $z'$ plus a variable independent of $z'$, and $z'$ is uniform on a cell, so each endpoint is uniform modulo the cell width $2^{-k}$ and the probability of a boundary in a window of length $d$ is $2^k d$. Hence $|\delta_P - \delta| \leq 2^{-s} + 2^{k+1} d < 2^{-s} + 2^{k+1-P}$. □

So the implementation is exact in the sense of (a), exact to within $2^{-s-2}$ in total variation by (b), and its failure probability is that of the idealized scheme to within $2^{-s} + 2^{k+1-P}$. The price is a key of $P \geq \log(1/q_{\min}) + s + 1$ bits, which can exceed the payload several times over (Section 6.4); it comes from asking for an exact guarantee on every sequence, including the least probable. The converse is stated for exact security. For a scheme that is secure only up to total variation $\varepsilon$, the same argument bounds $k$ by $H(\tilde{X}^n \mid W^n) + 1$, which differs from $n\Lambda$ by an amount that vanishes with $\varepsilon$ by the continuity of entropy; we do not need this refinement.

**What the guarantee covers.** The converse holds for every scheme obeying (i) and (ii), whatever the model; the achievability is for the cover law $Q$ only. If the process that generates natural texts is richer than $Q$, an observer who knows the richer model can

tell an embedded text from a natural one. Section 6.4 shows this for real genes: natural genes are overdispersed relative to the genome-wide synonymous law, and embedded genes are not. The scheme is the arithmetic-decoding construction used in provably secure steganography [10, 30, 49, 67]; what the theorem adds is the statement of the capacity in the ledger's terms, the exact failure probability, and the identification of what limits it: $\Lambda$, and nothing else.

**Scope of "meaning-preserving".** Condition (i) holds under the meaning map $\pi$ of the source. In the genetic code $\pi$ is translation, and the embedding is *protein-preserving*: the amino-acid sequence is unchanged. It is not claimed to be biologically neutral. Synonymous codons differ in translation speed, in mRNA structure and in their effects on expression and folding, and synonymous changes can alter protein function [34, 47]. For such purposes the meaning of a codon is finer than its amino acid (Theorem 4(d)), the latitude relative to it is smaller, and the capacity of the channel falls to that smaller latitude. Biological instances of the idea are old: messages have been written into DNA [12] and into the synonymous codons of genes [2].

### 4.3 Lossy meaning

If the receiver tolerates errors in meaning, the problem is one of *indirect* rate–distortion coding: reproduce a function of the source under a distortion measure that sees only that function. It was solved by Dobrushin and Tsybakov [17], Wolf and Ziv [60] and Witsenhausen [58]; the reduction to an ordinary rate–distortion problem with the distortion averaged over the belief, and its use for semantic sources [37], need no change here. The ledger adds nothing to that theory and we do not repeat it.

## 5 Estimating the ledger

Every quantity so far is a functional of a distribution. In practice one has a sample. This section states what can be said about the ledger of a source from $n$ independent observations of $(W, X)$, and what cannot.

### 5.1 The plug-in ledger

Replace $P$ by the empirical law $\hat{P}$. The five entries of $\hat{P}$ satisfy every identity of Theorem 1 exactly, since they are identities for any joint law. In particular the estimated balance $\hat{E} - \hat{C} = \hat{\Lambda} - \hat{\Delta}$ holds to rounding error. The estimates are not unbiased. By Jensen's inequality $\mathrm{E}[H(\hat{P}_Z)] \le H(P_Z)$ for any variable $Z$ with a finite range: the plug-in entropy is biased downward, and the bias is large when the alphabet is large relative to $n$ [39, 44]. Latitude and doubt are conditional entropies, and both are computed within strata, meanings for $\Lambda$ and forms for $\Delta$, that contain few observations when the alphabet is large. Their plug-in values are therefore too small.

Two corrections are in wide use. The Miller–Madow correction adds $(K_{\mathrm{obs}} - 1)/(2n \ln 2)$ bits to each entropy computed from $K_{\mathrm{obs}}$ observed categories [39]. The Chao–Shen estimator adjusts for the probability mass of unseen categories using the number of singletons [11]. For latitude and doubt we apply Miller–Madow within each stratum; Figure 3 compares the three estimators on a law of known truth.

**Table 2.** Certified half-width of Proposition 10 for $K = 25$ categories with probabilities proportional to $1/k$, at $\delta = 0.05$, against the observed error over 2000 replicates of each sample size.

| $n$ | $\varepsilon$ | Certified half-width (bits) | 95th percentile of actual error | Coverage |
|---|---|---|---|---|
| 500 | 0.285 | 1.840 | 0.158 | 1.000 |
| 1,000 | 0.202 | 1.402 | 0.103 | 1.000 |
| 2,000 | 0.143 | 1.063 | 0.067 | 1.000 |
| 5,000 | 0.090 | 0.732 | 0.042 | 1.000 |
| 10,000 | 0.064 | 0.549 | 0.030 | 1.000 |
| 30,000 | 0.037 | 0.346 | 0.017 | 1.000 |
| 100,000 | 0.020 | 0.207 | 0.009 | 1.000 |

### 5.2 A certified interval for a small alphabet

For a variable with a small range one can bound the error rigorously. This covers, for example, a coarse meaning such as a chemical class or a semantic field.

**Proposition 10** (Certified interval)**.** *Let $Z$ have a range of $K \geq 2$ values and let $\hat{p}$ be the empirical law of $n$ independent copies. For $\delta \in (0,1)$ put $\varepsilon = \sqrt{(2/n)\,(K\ln 2 + \ln(1/\delta))}$. If $\varepsilon \leq \frac{1}{2}$, then with probability at least $1-\delta$*

$$\left|H(\hat{p}) - H(p)\right| \leq \varepsilon \log \frac{K}{\varepsilon}.$$

*Proof.* For the empirical law of $n$ independent draws from a distribution on $K$ points, $\mathbb{P}[\|\hat{p} - p\|_1 \geq \varepsilon] \leq 2^K e^{-n\varepsilon^2/2}$; this is the inequality of Bretagnolle and Huber-Carol in the form given by Weissman et al. [57], whose constant is $(2^K - 2)$. Setting the right side equal to $\delta$ and solving gives the stated $\varepsilon$, so $\|\hat{p} - p\|_1 \leq \varepsilon$ with probability at least $1-\delta$. If $\|p - q\|_1 = \eta \leq \frac{1}{2}$, then $|H(p) - H(q)| \leq -\eta \log(\eta/K)$ [14, Thm. 17.3.3]. The function $\eta \mapsto \eta \log(K/\eta)$ is increasing on $(0, K/e)$, and $\frac{1}{2} < K/e$ for $K \geq 2$, so the bound at $\eta \leq \varepsilon$ is at most its value at $\varepsilon$. □

The interval needs $n \geq 8\,(K\ln 2 + \ln(1/\delta))$ to exist, and it is conservative. Table 2 gives the half-width beside the observed error for $K = 25$: it never failed to cover in 2000 replicates per size, and it is 12–22 times wider than the observed 95th percentile of the error. It is a guarantee, not an estimate. It applies to the entropy of a small alphabet, so it certifies $C$ when the meanings are coarse; it does not certify $E$, $\Lambda$ or $\Delta$ for a large vocabulary.

### 5.3 Validation on a law of known truth

We built a joint law with the structure the ledger must handle: 3000 meanings and 2500 forms, both with Zipf–Mandelbrot weights, up to four forms per meaning, forms chosen in proportion to the square root of their frequency so that frequent forms are more ambiguous, and Dirichlet weights inside each meaning. Its ledger is computed exactly: $\Lambda = 0.275$ bits and $\Delta = 0.625$ bits. We drew 60 samples at each size from $2 \times 10^3$ to $3 \times 10^5$ and applied the three estimators. Figure 3 shows the bias. At $n = 2000$ the plug-in estimates of latitude and doubt are too small by $-0.110$ and $-0.339$ bits, that is by 40 and 54 %. Miller–Madow removes most of this at large $n$ ($-0.003$ and $-0.003$ bits at $n = 10^5$). Chao–Shen overshoots at intermediate sizes. All three converge as $n$ grows.

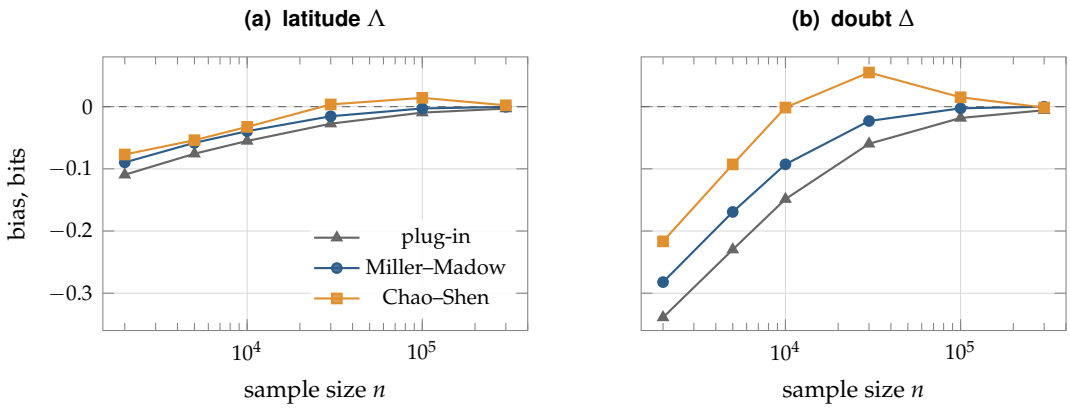



**Figure 3.** Bias of three estimators of latitude and doubt on a Zipf-type law with known truth ($\Lambda = 0.275$ bits, $\Delta = 0.625$ bits; 60 replicates per size). All three underestimate at small $n$; from $n = 100{,}000$ on, Miller–Madow has the smallest absolute bias for both.

**Table 3.** The plug-in ledger of SemCor as the sample grows (mean of 12 random subsamples; the last row is the whole corpus). Bits. The values of $C$ and $E$ are still rising at the full corpus.

| Fraction of corpus | Tokens | $C$ | $E$ | $\Lambda$ | $\Delta$ |
|---:|---:|---:|---:|---:|---:|
| 2 % | 3,679 | 10.55 | 10.21 | 0.126 | 0.464 |
| 5 % | 9,199 | 11.24 | 10.80 | 0.188 | 0.629 |
| 10 % | 18,398 | 11.64 | 11.11 | 0.234 | 0.761 |
| 25 % | 45,996 | 12.02 | 11.41 | 0.299 | 0.911 |
| 50 % | 91,993 | 12.20 | 11.55 | 0.340 | 0.992 |
| 100 % | 183,986 | 12.33 | 11.65 | 0.375 | 1.051 |

### 5.4 What a finite corpus does and does not determine

For a distribution with finite entropy the plug-in estimator is strongly consistent, but there is no universal rate: for every sequence $a_n \to 0$ there are distributions whose error exceeds $a_n$ infinitely often [1, 53]. When the alphabet is large relative to $n$ the plug-in estimator is also far from minimax-optimal [44, 62]. A statement about how well a sample reveals a population entropy therefore needs the sample size, the alphabet and the tail, and we make none for heavy-tailed laws in general. The validation law above has finite entropy, and all three estimators converge on it (Figure 3).

For SemCor we make only statements about the sample. Table 3 shows that the plug-in values of $C$ and $E$ are still rising at the full corpus: between half the corpus and all of it, $C$ rises by 0.12 bits and $E$ by 0.10 bits. The whole-corpus values are therefore not converged at this size, and we claim no value for a larger population. What we report instead are quantities that are stable under the checks of Section 6.5: the ledger of a well-attested core, obtained by keeping only forms and meanings with at least $\tau$ observations; conditional profiles of well-attested items; and the sign of the balance $\Delta - \Lambda$, together with how that sign changes when the definition of the form changes. Each time, the estimator, the restriction of the source and the resampling unit are stated.

**Table 4.** Coding sequences used, with the filters applied.

| Genome | Accession | CDS in file | Excluded | Used |
|---|---|---|---|---|
| E. coli K-12 | U00096.3 | 4,298 | 3 | 4,295 |
| S. aureus NCTC 8325 | CP000253.1 | 2,892 | 0 | 2,892 |
| S. argenteus | FR821777.2 | 2,510 | 0 | 2,510 |
| S. agnetis | CP045927.1 | 2,299 | 0 | 2,299 |
| C. simulans PES1 | NZ_CP014634.1 | 2,566 | 0 | 2,566 |
| A. thaliana plastid | NC_000932.1 | 85 | 0 | 85 |

# 6 Measurements

The theory is exact, so measurements cannot confirm it. They serve three other purposes: to show the size of each term in real systems, to test the estimation protocol of Section 5, and to show that the identities are usable, that is, that quantities computed from one part of the data predict another. We use two systems, the genetic code and sense-tagged English, chosen because in each the meaning of a message is fixed by something other than the analyst.

## 6.1 Data and meaning maps

**Genomes.** We use the coding sequences of six genomes: *Escherichia coli* K-12 MG1655 [8], *Staphylococcus aureus* NCTC 8325, *S. argenteus*, *S. agnetis*, *Corynebacterium simulans* PES1, and the plastid of *Arabidopsis thaliana*. They span a range of base composition (GC content of coding sequence from 0.33 to 0.60) and include a closely related trio, which serves as a hard case. Sequences are read from the annotated coding sequences of public GenBank records; Table 4 lists accessions and the records excluded (a coding sequence is excluded if its length is not a multiple of three, if it contains a character outside `ACGT`, or if it has an in-frame stop codon before its end; the three excluded *E. coli* records are the selenoprotein genes, in which UGA is recoded). Every gene contributes its codons and its final stop codon. The provenance of each file is in Section A.

**Meaning maps.** $X$ is the codon (64 values). $W$ is the amino acid or stop (21 values) by the standard genetic code; the bacterial and plastid code differ from it only in initiation codons. $\Delta = 0$. As a coarser level we use the six Dayhoff groups of amino acids [16], cysteine; alanine, glycine, proline, serine, threonine; aspartate, glutamate, asparagine, glutamine; histidine, lysine, arginine; isoleucine, leucine, methionine, valine; phenylalanine, tryptophan, tyrosine, together with stop, giving $G$ with seven values.

**Purposes.** $Y_1$ is the genome a codon came from, uniform over the six. $Y_2$ is whether a codon belongs to a highly expressed *E. coli* gene: a list of 253 such genes is distributed with a codon-design tool, and 251 of them match, by exact sequence, a gene among the 4,295 usable ones; they contribute 6.5 % of all codons.

**Uncertainty.** In this section all genome intervals are 95 % percentile intervals over 400 resamplings of genes with replacement, since genes and not codons are the independent units; the intervals for SemCor are described in Section 6.5.

**Table 5.** The ledger of the genetic code, $\Delta = 0$. Bits per codon; GC3 is the fraction of G or C at the third position of sense codons; $2^E$ is the effective number of codons.

| Genome | Genes | Codons | GC3 | $E$ | $2^E$ | $C$ | $\Lambda$ | $\Lambda/E$ |
|---|---|---|---|---|---|---|---|---|
| E. coli K-12 | 4,295 | 1,331,529 | 0.56 | 5.676 | 51.1 | 4.175 | 1.501 | 26.5 % |
| S. aureus NCTC 8325 | 2,892 | 799,920 | 0.23 | 5.423 | 42.9 | 4.163 | 1.260 | 23.2 % |
| S. argenteus | 2,510 | 770,437 | 0.21 | 5.394 | 42.1 | 4.160 | 1.235 | 22.9 % |
| S. agnetis | 2,299 | 697,903 | 0.27 | 5.510 | 45.6 | 4.175 | 1.335 | 24.2 % |
| C. simulans PES1 | 2,566 | 821,485 | 0.71 | 5.576 | 47.7 | 4.112 | 1.465 | 26.3 % |
| A. thaliana plastid | 85 | 26,494 | 0.29 | 5.661 | 50.6 | 4.187 | 1.474 | 26.0 % |

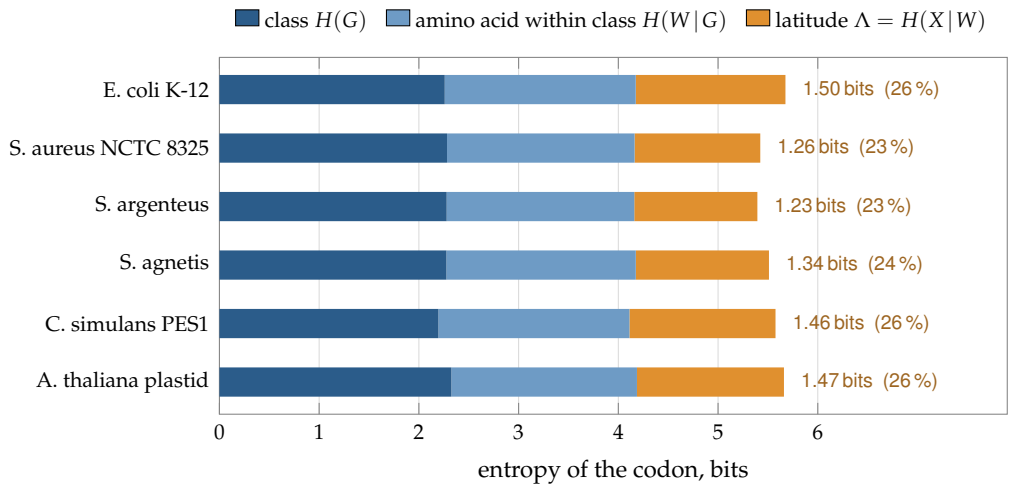



**Figure 4.** The entropy of the codon in six genomes, split over three levels of meaning: the chemical class of the amino acid (Dayhoff groups and stop), the amino acid within its class, and the synonymous codon within its amino acid (latitude). The label at the end of each bar gives latitude in bits and as a share of $E = H(X)$.

### 6.2 The ledger of the genetic code

Table 5 and Figure 4 give the ledger of each genome. The latitude lies between 1.235 and 1.501 bits per codon, which is 22.9–26.5 % of the entropy of the codon. The effective vocabulary $2^E$ is 42.1–51.1 of the 64 codons (66–80 %), so Corollary 2 is satisfied with room to spare. The genomes differ far more in how they say than in what they say: the content $C$, the entropy of the protein alphabet, varies by 0.08 bits over the six genomes and the latitude by 0.27 bits. The gene-resampling intervals for $\Lambda$ have half-width at most 0.01 bits for the five bacterial genomes and 0.035 for the plastid, which has only 85 genes.

### 6.3 Where identity lives

**The panel.** Let $Y_1$ be uniform on the six genomes and draw a codon from the chosen genome. The codon carries $I(X;Y_1) = 0.178$ bits about the genome, out of $\log_2 6 = 2.585$ bits. Table 6 splits this over the three levels of Proposition 6. The chemical class carries 0.0060 bits, the amino acid within its class 0.0146 bits, and the choice among synonymous codons 0.157 bits: 88.5 % of the total (interval 87.4–89.2 %). The genetic code is far from adequate for the purpose of identifying the genome: its leakage is the greater part of the information. In these genomes the greater part of the information about origin sits in the latitude, the part of the message that the genetic code leaves free.

**Table 6.** Where the information about the genome lives (Proposition 6). The three terms sum to $I(X;Y_1)$; intervals from resampling genes.

| Level of meaning | Term | Bits per codon | 95% interval | Share |
|---|---|---|---|---|
| Chemical class ($G$) | $I(G;Y)$ | 0.0060 | [0.0054, 0.0069] | 3.4 % |
| Amino acid within its class | $I(W;Y \mid G)$ | 0.0146 | [0.0135, 0.0163] | 8.2 % |
| Synonymous choice: leakage | $I(X;Y \mid W)$ | 0.1573 | [0.1558, 0.1598] | 88.5 % |
| Codon (total) | $I(X;Y)$ | 0.1779 | [0.1759, 0.1812] | 100.0 % |

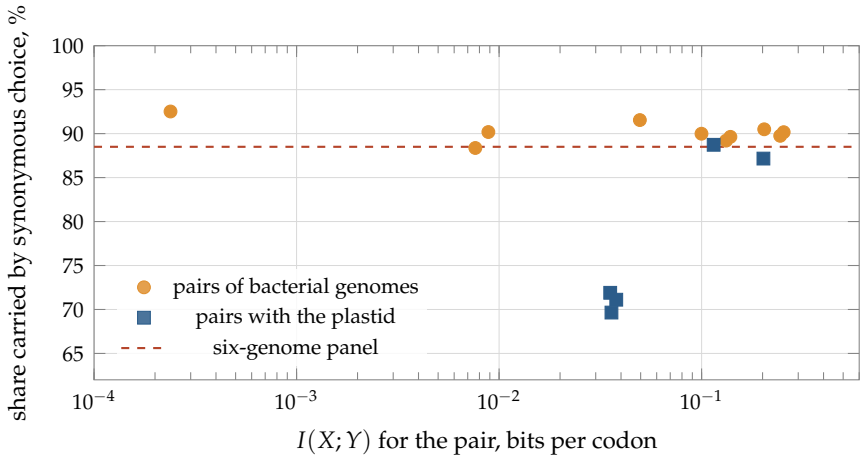



**Figure 5.** For each of the 15 pairs of genomes, the share of the information that identifies the genome of a codon that is carried by the choice among synonymous codons, $I(X;Y \mid W)/I(X;Y)$, against the total information of the pair. The share is stable over three orders of magnitude of $I(X;Y)$; it is lower when the plastid is one member of the pair, because its amino-acid composition differs more.

**Pairs.** Figure 5 repeats the split for each of the 15 pairs. For the ten pairs of bacterial genomes the share is between 88 and 93 %, over three orders of magnitude of total information (the leftmost point is the pair *S. aureus* and *S. argenteus*, whose codon usage differs by only 0.0002 bits). Pairs that contain the plastid lie between 70 and 89 %: organelle genes differ from bacterial genes in amino-acid composition as well.

**A held-out test.** The chain rule for relevance is an identity, so a test of it must be a prediction. We split the genes of each genome into two halves at random, fitted the three class-conditional laws $P_y(x)$, $P_y(w)$ and $P_y(x \mid w)$ on one half with a smoothing constant of $\frac{1}{2}$, and on the other half (i) assigned each gene to the genome with the highest log-likelihood under each law, and (ii) measured the information gain $\mathrm{E}[\log P_y(x) - \log \bar{P}(x)]$ per codon, averaged over the six genomes, where $\bar{P}$ is the equal mixture of the six laws. The last quantity should equal the training-half information; for the synonymous term it is defined by the chain rule as the codon gain minus the amino-acid gain. Table 7 shows that it does. The held-out gain for codons is 0.1773 bits against 0.1774 on the training half; for amino acids 0.0189 against 0.0215; for the synonymous choice 0.1583 against 0.1559. The largest discrepancy is 0.003 bits. As classifiers, the synonymous choice alone identifies the genome of a gene almost as well as the full codon sequence (macro recall 0.735 against 0.740), and the amino acids alone identify it far less well (0.469; chance is

**Table 7.** Held-out test of the relevance ledger on 7,325 genes: information per codon under the law fitted on the other half, and identification of the genome of a gene by the evidence shown.

| | Information, bits per codon | | Identification |
|---|---|---|---|
| Evidence used | Training law | Held-out gain | macro recall |
| Codons | 0.1774 | 0.1773 | 0.740 |
| Amino acids only | 0.0215 | 0.0189 | 0.469 |
| Synonymous choice only | 0.1559 | 0.1583 | 0.735 |

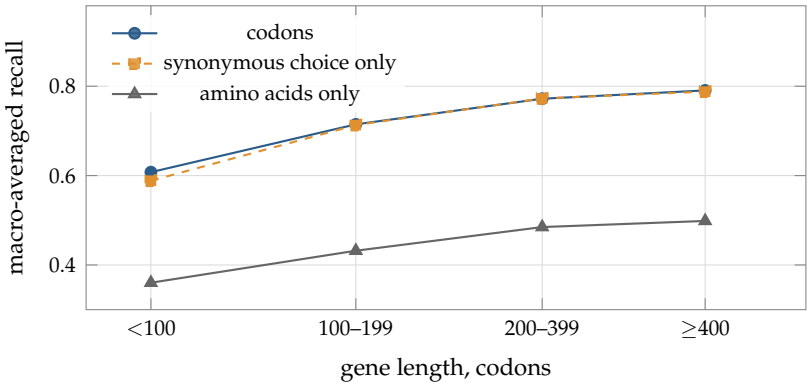



**Figure 6.** Held-out identification of the genome of a gene, by the evidence used, against gene length. Six classes; chance is $1/6 \approx 0.17$. The synonymous choice alone almost matches the full codon sequence; the protein alone falls far short.

0.17). Figure 6 shows the dependence on gene length. Most of the errors of the codon classifier are between *S. aureus* and *S. argenteus*, which are nearly indistinguishable by codon usage.

**A second purpose.** The same genes carry information about a different question: is this gene highly expressed? For $Y_2$, $I(X;Y_2) = 0.0131$ bits per codon of the 0.347 bits of uncertainty in $Y_2$. The amino acid carries 0.0013 bits and the synonymous choice 0.0118 bits, so 90.2 % (interval 88.6–91.3 %) of what a codon says about expression is in the choice among synonyms. This is the translational-selection effect in highly expressed bacterial genes [26, 52], stated as a leakage: for the purpose "how much protein is made", the genetic code is again far from adequate. Two purposes give two leakages on one source; this is Theorem 5(d) at work, and it is why the meaning of a codon cannot be stated without a purpose.

### 6.4 Latitude as a channel, on real genes

To test Theorem 8 and Proposition 9 we took 240 randomly chosen *E. coli* genes of 60 to 700 codons. For each gene the cover meaning $w^n$ is its amino-acid sequence, the final stop codon included, and the cover law is the genome-wide synonymous law $P(x \mid a)$ of *E. coli*, kept as exact fractions of the codon counts (assumptions A1 and A2). We used the scheme of the proof with exact rational arithmetic. The payload had $k = \lfloor \mu - 3\sigma - 7 \rfloor$ bits for each gene ($\mu$ and $\sigma$ as in the theorem). The key $R$ and the dither $V$ had $P = \lceil \sum_i \max_x(-\log Q(x \mid a_i)) \rceil + 65$ bits, which satisfies the condition of Proposition 9 with $s = 64$: the law of the embedded text is within total variation $2^{-66}$ of the cover law, and the

**Table 8.** Embedding a hidden message in the synonymous choices of real *E. coli* genes. Capacity is $\sum_i H(X \mid a_i)/n$, the exact expected number of bits per codon; the rate is the payload actually used.

| Length (codons) | Genes | Capacity, bits/codon | Rate $k/n$, bits/codon | Decoded correctly |
|---|---|---|---|---|
| 60–119 | 30 | 1.437 | 1.126 | 100 % |
| 120–199 | 53 | 1.461 | 1.258 | 100 % |
| 200–349 | 88 | 1.503 | 1.355 | 100 % |
| $\geq 350$ | 69 | 1.499 | 1.391 | 100 % |
| All | 240 | 1.494 | 1.351 | 100 % |

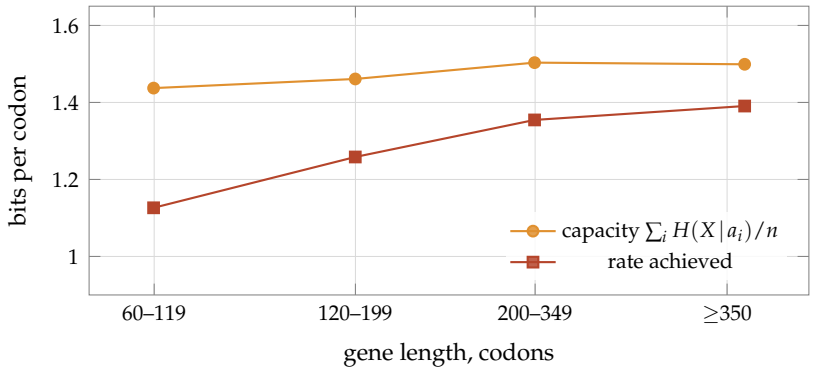



**Figure 7.** Bits hidden per codon in real *E. coli* genes without changing the encoded amino-acid sequence: the capacity of the synonymous choice (Theorem 8) and the rate actually achieved with a safety margin of three standard deviations. The gap closes as $1/\sqrt{n}$.

failure probability is within $2^{-63}$ of the idealized one. The mean key length was 817 bits, 2.2 times the payload. The random bits came from a seeded Mersenne Twister generator so that the experiment is reproducible; this is not a secret key in the sense of A3, and a deployment would need a cryptographic generator. We decoded each payload from the codons, the amino-acid sequence and the key, and checked that the embedded gene translates to the natural protein.

240 of the 240 genes decoded correctly; 0 failures were reported and 0 wrong messages were returned. Every embedded gene translates to the same amino-acid sequence as the natural one. In total 90,127 bits were hidden in 66,694 codons, a rate of 1.351 bits per codon against a capacity $\sum_i H(X \mid a_i)/n$ of 1.494 (90.5 % of capacity); Table 8 and Figure 7 show the rate approaching capacity as genes lengthen. On average 58 % of the codons of a gene differ from its natural ones. As a demonstration, a sentence of 51 bytes with a 16-bit tag (424 bits; capacity of the gene 475 bits) was written into a gene of 292 codons, changing 181 of them and no amino acid; it decoded correctly after 1 attempt (a detected failure would be retried with fresh randomness). Figure 8 shows the first codons.

These genes are protein-preserving, not shown to be biologically neutral: we make no claim about expression, folding or fitness (Section 4.2). The security claim is relative to the cover law and is testable. Pooled over the 240 genes, the codon counts of the embedded genes are consistent with the law $P(x \mid a)$ (Pearson $\chi^2$ with 43 degrees of freedom, $p = 0.20$), and those of the natural genes are not ($p = 0.006$). Per gene, $\chi^2$ divided by its degrees of freedom is 0.99 for the embedded genes and 1.59 for the natural ones. Natural genes are overdispersed relative to the genome-wide law, because codon bias varies from gene to gene, and embedded genes are not. An observer who models this

```
natural   ATG ACC ACA CTG GCG ATT GAT ATC GGC GGT ACT AAA CTT GCC GCC GCG
embedded  ATG ACT ACG CTG GCC ATC GAT ATT GGT GGT ACC AAA CTT GCG GCG GCG
```

**Figure 8.** The first 16 codons of the gene of the demonstration, natural above and with the message embedded below. Codons that differ are in bold red; both rows translate to the same amino-acid sequence.

variation could tell the two apart, as Section 4.2 warned; the guarantee is for the cover law used.

### 6.5 English: synonymy and ambiguity together

In natural language neither map is a function. We use SemCor, the portion of the Brown Corpus in which each content word carries its WordNet sense [19, 40, 41], as distributed with NLTK [6]. We take the 186 documents in which all content words are tagged, giving 183,986 tokens, each with a lemma ($X$, part-of-speech blind, 19,658 types) and a synset ($W$, 24,672 types). Named-entity chunks without a synset are excluded. The main analysis takes the lemma as the form, so inflection lies outside the latitude; the sensitivity analysis below changes this.

**Estimation.** The plug-in ledger of the whole corpus has $\Lambda = 0.375$ and $\Delta = 1.051$ bits (Miller–Madow 0.40 and 1.10), so $E < C$ (11.65 against 12.33 bits). As Table 3 showed, these numbers still move with the sample size, so we restrict attention to the well-attested core: for a threshold $\tau$ we repeatedly delete every lemma and every synset with fewer than $\tau$ tokens until none remains, and compute the ledger of what is left. Intervals are 95 % normal intervals from the bootstrap standard error over 1,000 resamplings of documents, the independent unit. We did not use percentile intervals, because a resample contains fewer distinct types than the sample and diversity-type statistics are shifted by that. For the whole corpus the bootstrap standard error of $\Delta - \Lambda$ (0.010 bits) agrees with half-sampling (0.010) and with the delete-one-document jackknife (0.010). In the $\tau = 20$ core the jackknife is 1.5 times larger than the bootstrap (0.021 against 0.014 bits), because the core changes with the resample; the intervals there may be somewhat too narrow, but no conclusion below depends on this.

**The ledger of the core.** Table 9 and Figure 10(a) give $\Lambda$, $\Delta$ and their difference. For $\tau = 20$, which keeps 53 % of the tokens, $\Lambda = 0.26$ and $\Delta = 0.71$ bits per word, so $\Delta - \Lambda = 0.44 \pm 0.03$ bits; the plug-in and Miller–Madow values differ by at most 0.003 bits for $\Lambda$ and 0.005 bits for $\Delta$ from $\tau = 20$ upward. For the whole corpus $\Delta - \Lambda = 0.70 \pm 0.02$. The balance is positive at every $\tau$ in the table.

**Sensitivity.** Table 10 and Figure 9 change what the analysis calls a form, the part of speech and the collection, at $\tau = 20$ and, for the balance, at $\tau = 10$.

*Form.* With the part of speech added to the lemma, the balance is $0.37 \pm 0.03$: knowing the category removes doubt between categories (a noun and a verb with one spelling) but not within them. With the inflected word as written, the balance is $-0.20 \pm 0.05$: latitude rises to 0.96 bits because inflection (*took*, *taken*, *taking*) is freedom that the synset does not fix, and doubt changes little, so latitude now exceeds doubt. The sign of $E - C$ therefore depends on whether grammatical variation counts as part of the form.

**Table 9.** The ledger of the well-attested core of SemCor, form: lemma. $\tau$ is the minimum number of tokens of every kept lemma and synset. Miller–Madow estimates in bits with 95 % intervals from 1,000 document resamplings.

| $\tau$ | Tokens kept | Fraction kept | $\Lambda$ | $\Delta$ | $\Delta - \Lambda$ |
|---|---|---|---|---|---|
| 1 | 183,986 | 100 % | $0.400 \pm 0.010$ | $1.095 \pm 0.023$ | $0.695 \pm 0.020$ |
| 2 | 171,183 | 93 % | $0.377 \pm 0.010$ | $1.063 \pm 0.024$ | $0.686 \pm 0.021$ |
| 3 | 161,453 | 88 % | $0.360 \pm 0.009$ | $1.026 \pm 0.024$ | $0.666 \pm 0.022$ |
| 5 | 147,382 | 80 % | $0.340 \pm 0.009$ | $0.961 \pm 0.023$ | $0.621 \pm 0.022$ |
| 10 | 123,392 | 67 % | $0.302 \pm 0.011$ | $0.834 \pm 0.022$ | $0.532 \pm 0.022$ |
| 20 | 97,869 | 53 % | $0.264 \pm 0.012$ | $0.705 \pm 0.026$ | $0.442 \pm 0.028$ |
| 50 | 63,643 | 35 % | $0.171 \pm 0.018$ | $0.547 \pm 0.030$ | $0.376 \pm 0.034$ |

**Table 10.** Sensitivity of the SemCor ledger to the definition of the form, the part of speech and the collection. Bits; Miller–Madow estimates with 95 % intervals from 1,000 document resamplings. "Kept" is the fraction of the tokens of the analysis that survive the core restriction.

| | | $\tau = 20$ | | | $\tau = 10$ |
|---|---|---|---|---|---|
| Analysis | Kept | $\Lambda$ | $\Delta$ | $\Delta - \Lambda$ | $\Delta - \Lambda$ |
| *Form of the word* | | | | | |
| Lemma (main analysis) | 53 % | $0.26 \pm 0.01$ | $0.71 \pm 0.03$ | $0.44 \pm 0.03$ | $0.53 \pm 0.02$ |
| Lemma and part of speech | 53 % | $0.24 \pm 0.01$ | $0.61 \pm 0.02$ | $0.37 \pm 0.03$ | $0.41 \pm 0.02$ |
| Inflected word as written | 43 % | $0.96 \pm 0.06$ | $0.76 \pm 0.03$ | $-0.20 \pm 0.05$ | $-0.23 \pm 0.04$ |
| *Part of speech (form: lemma)* | | | | | |
| Nouns | 48 % | $0.15 \pm 0.02$ | $0.38 \pm 0.03$ | $0.23 \pm 0.03$ | $0.30 \pm 0.02$ |
| Verbs | 59 % | $0.36 \pm 0.02$ | $1.26 \pm 0.04$ | $0.90 \pm 0.05$ | $0.93 \pm 0.04$ |
| Adjectives | 46 % | $0.22 \pm 0.03$ | $0.22 \pm 0.03$ | $0.00 \pm 0.04$ | $0.07 \pm 0.04$ |
| Adverbs | 70 % | $0.33 \pm 0.03$ | $0.43 \pm 0.04$ | $0.10 \pm 0.05$ | $0.09 \pm 0.04$ |
| *Collection (form: lemma)* | | | | | |
| Brown 1 (103 documents) | 43 % | $0.20 \pm 0.02$ | $0.56 \pm 0.04$ | $0.35 \pm 0.04$ | $0.43 \pm 0.03$ |
| Brown 2 (83 documents) | 40 % | $0.16 \pm 0.02$ | $0.56 \pm 0.04$ | $0.40 \pm 0.04$ | $0.43 \pm 0.04$ |

*Part of speech.* Doubt exceeds latitude for nouns ($0.23 \pm 0.03$), verbs ($0.90 \pm 0.05$) and adverbs ($0.10 \pm 0.05$); the two are balanced for adjectives ($0.00 \pm 0.04$). Verbs account for most of the doubt.

*Collection.* The two collections of SemCor, analysed separately, agree: 0.35 and 0.40 at $\tau = 20$, and 0.43 and 0.43 at $\tau = 10$.

**Profiles.** Figure 10(b) breaks the ledger down by item. Among the 1,685 lemmas with at least 20 tokens, doubt rises with frequency, from 1.08 bits (about 2.1 effective senses) in the lowest band to 1.72 bits (about 3.3 senses) for lemmas with 160–319 tokens; the top band, with 35 lemmas of at least 320 tokens, is slightly lower at 1.60 bits. The Spearman correlation with the logarithm of the count is 0.21. This is Zipf's meaning-frequency relation [68] in the ledger's units. The most ambiguous frequent lemmas are *take* (4.5 bits), *draw*, *break* and *line*. Latitude per synset rises only slowly, from 0.46 to 0.57 bits over the

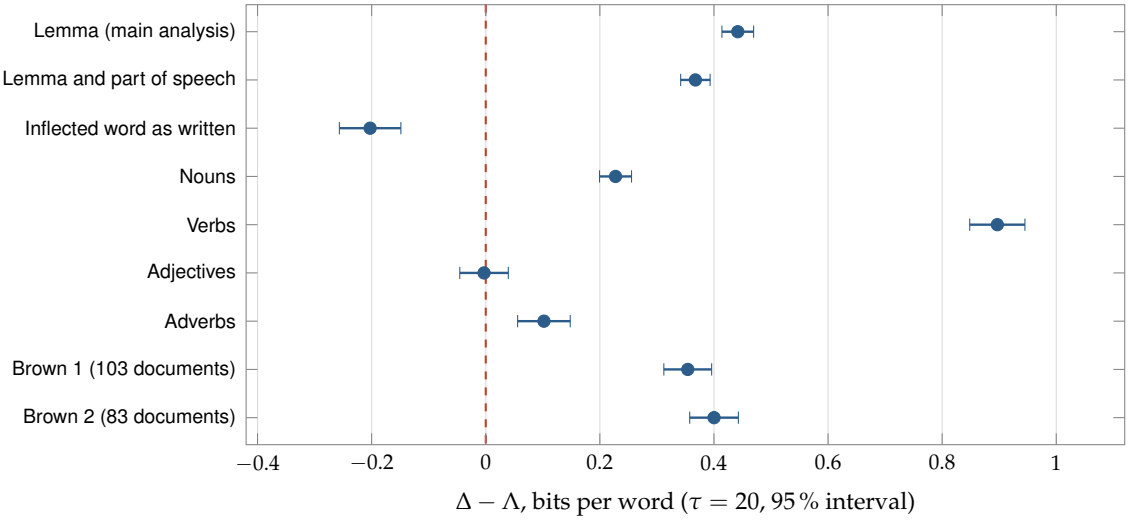



**Figure 9.** Sensitivity of the sign of the balance $E - C = \Lambda - \Delta$ in SemCor. Each row is one analysis of Table 10; the dashed line is $\Delta = \Lambda$. Doubt exceeds latitude for lemmas, for lemmas with their part of speech, for nouns, verbs and adverbs, and in both collections; latitude exceeds doubt when inflected words are the forms; adjectives are balanced.

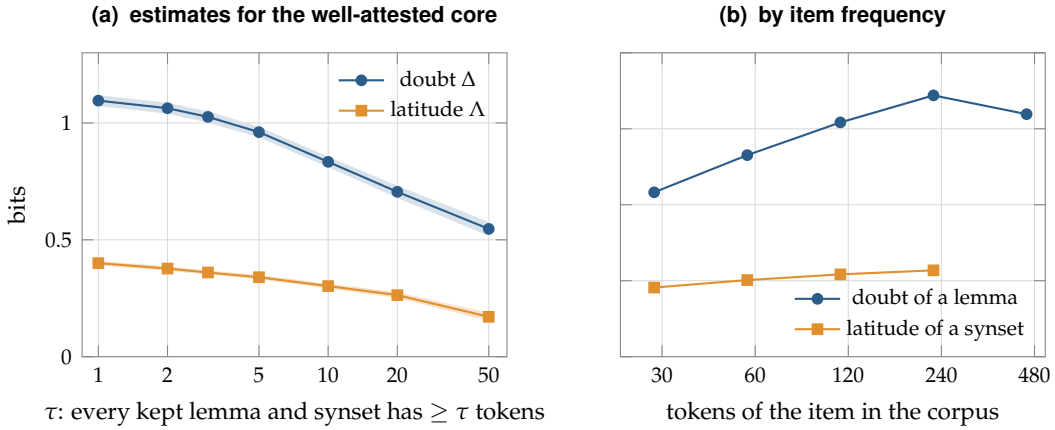



**Figure 10.** The ledger of English content words in SemCor. (a) Doubt and latitude, Miller–Madow estimates with 95 % intervals from resampling documents, as the source is restricted to items with at least $\tau$ tokens. (b) Mean doubt per lemma and mean latitude per synset, among items with at least 20 tokens, in frequency bands. Doubt rises with frequency; latitude rises only slowly.

same range of counts, with a rank correlation of 0.07.

### 6.6 A sense inventory without a corpus

The corpus ledger weights each word by how often it is used. A sense inventory has no usage. For a wordnet we count every distinct pair (synset, lemma) once, so that $P(w, x)$ is proportional to 1 on the pairs; the lemma is the lower-cased string, as in the main SemCor analysis, so a spelling shared by a noun and a verb is one lemma. Then $\Lambda$ is the mean over pairs of $\log_2$ of the number of lemmas in the pair's synset, $\Delta$ is the mean over pairs of $\log_2$ of the number of synsets of the pair's lemma, and the balance of Theorem 1 holds exactly (the code checks it). English WordNet 3.0 has 206,941 pairs in 117,659 synsets, with $\Lambda = 1.05$ and $\Delta = 0.83$ bits: here latitude exceeds doubt, the opposite sign from the lemmas of SemCor. The two analyses differ in weighting, and Figure 10(b) shows that

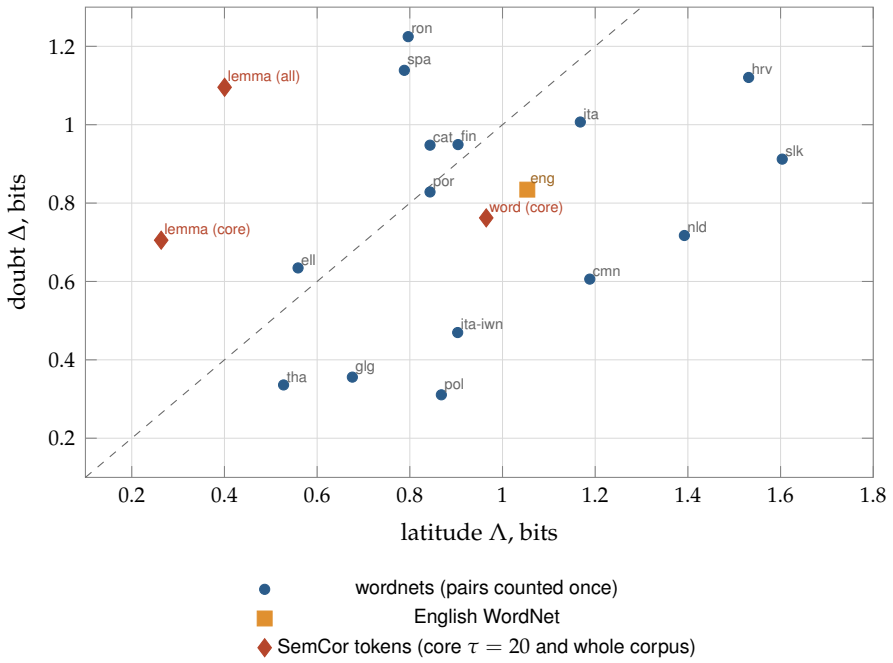



**Figure 11.** Doubt against latitude. Blue: the 22 wordnets with at least 10,000 synsets, each distinct (synset, lemma) pair counted once; orange: English WordNet 3.0; red diamonds: SemCor tokens, weighted by use (lemma, whole corpus; lemma and inflected word, $\tau = 20$ core). Points above the dashed line have doubt greater than latitude. The balance has no fixed sign in the inventories, and flips between the lemma and the inflected word in the corpus.

doubt rises with frequency, which is consistent with weighting by use moving the balance toward doubt; we have not isolated this as the cause.

The Open Multilingual Wordnet [9] attaches wordnets of other languages to the same synsets. Among the 22 wordnets with at least 10,000 synsets (of 32 in the package; Table 12 lists all), latitude exceeds doubt in 12 and doubt exceeds latitude in 10 (Figure 11). The difference $\Lambda - \Delta$ ranges from $-0.62$ to 1.23 bits. These numbers describe the resource as much as the language, since wordnets are built by different methods and with different policies for synonyms. We draw one conclusion: neither sign of the balance is a general property of lexicons, and the corpus result is a statement about English usage under a stated definition of the form.

### 6.7 Verification of the identities

Every identity and theorem of Sections 2 to 4 that can be checked numerically was checked on random instances (Table 11). These checks do not support the theorems, which are proved; they guard against algebraic slips and coding errors, and they exercised the same library that computed the tables above. The minimality check enumerates every set partition of alphabets of three to six messages and confirms that each partition that loses no information is a refinement of the belief partition.

**Table 11.** Randomized verification of the identities and theorems (100,000 random joint laws for the first block).

| Check | Instances | Result |
|---|---|---|
| Ledger partitions $E = I + \Lambda$, $C = I + \Delta$, $H(W, X) = I + \Lambda + \Delta$ | 100,000 | max error $8.9 \times 10^{-16}$ |
| Balance $E - C = \Lambda - \Delta$ | 100,000 | max error $4.4 \times 10^{-16}$ |
| Vocabulary bound $\Delta \geq C - \log_2 N + \Lambda$ | 100,000 | 0 violations |
| Huffman length minus entropy, for $W$ and for $W \mid X$ | 3,000 | in $[0, 1)$ to within $3 \times 10^{-16}$ |
| Sufficiency $I(M; Y) = I(X; Y)$ | 600 | max error $1.8 \times 10^{-15}$ |
| Minimality; all set partitions of alphabets of 3 to 6 messages | 4,119 | each sufficient partition refines $M$ |
| Monotonicity in purpose | 404 | 404 of 404 hold |
| Leakage equals $I(X; Y) - I(W; Y)$ | 5,000 | max error $2.7 \times 10^{-15}$ |
| Depth identity $\mathcal{L}_Y(W_2) = \mathcal{L}_Y(W_1) + I(W_1; Y \mid W_2)$ | 5,000 | max error $3.6 \times 10^{-15}$ |
| Purpose $Y = X$ gives leakage $H(X \mid W) = \Lambda$ | 5,000 | max error $1.1 \times 10^{-15}$ |

## 7 Relation to earlier work

**Information and meaning.** Bar-Hillel and Carnap measured the semantic information of a statement by the possible worlds it excludes [4]; Floridi develops a philosophy of semantic information in which data must be well formed, meaningful and truthful [23]. Both give a measure of a statement, not of a source, and no coding theorem. The ledger is a measure of a source, obtained from a joint law, and every entry has a coding meaning. Kolchinsky and Wolpert [35] define semantic information as the information a system has about its environment that is causally necessary for its continued existence; the purpose $Y$ of Definition 3 plays the role of what matters to the receiver, but in our setting it is any variable the analyst names, and no physical theory of agency is used.

**Semantic communication.** The engineering literature on semantic and goal-oriented communication is large [3, 27]. Formulations that ask the receiver to reproduce a function of the source are indirect rate–distortion problems, whose theory is due to Dobrushin and Tsybakov [17], Wolf and Ziv [60] and Witsenhausen [58] and which Liu et al. [37] apply to semantic sources with an intrinsic and an extrinsic state. Goldreich et al. [25] treat communication in which meaning arises from the goal of the exchange and the parties have no common language, and Juba et al. [33] show that ambiguity can be efficient when the parties do not share a prior. Our results on coding are consequences of Shannon's theorems and add nothing to these theories; what is added is the accounting that separates the three quantities every such formulation uses, and the leakage that tests a chosen meaning.

**Meaning as sufficiency.** Defining the meaning of a message as the belief it induces is the construction of a minimal sufficient statistic [22], of the posterior as the canonical form of an experiment [7], of the information bottleneck at the point of no loss [55], and of the causal state of computational mechanics [50]. Distributional clustering of words by the contexts they occur in [45] and the distributional hypothesis behind it [21, 28] are the

same idea applied to language. What Theorem 4 and Theorem 5 add is bookkeeping: the three-term split of the entropy into consequence, hedge and latitude, the monotonicity in the purpose, the leakage of a declared meaning, and the statement that no meaning is adequate for every purpose.

**Ambiguity, synonymy and efficiency.** That languages trade the speaker's effort against the listener's is Zipf's principle of least effort [69]; Ferrer i Cancho and Solé [20] put it in information-theoretic form with a cost that mixes the entropy of the signals and the information they carry. Piantadosi et al. [46] argue that ambiguity is efficient because context disambiguates, and Gibson et al. [24] review the efficiency view of language, and Zaslavsky et al. [65] give an information-bottleneck account of colour naming. The vocabulary law of Corollary 2 is an accounting statement inside this tradition: it says how much vocabulary a given content and freedom require, without a claim about what speakers optimise.

**Semantic entropy of language models.** Kuhn et al. [36] and Farquhar et al. [18] cluster the answers of a language model by bidirectional entailment and take the entropy of the cluster distribution as an uncertainty score that flags likely confabulation. In the ledger's terms the clusters are a meaning map $\pi$ and their score is the content $H(\pi(X))$ of the model's output distribution; the ordinary sequence entropy is the expression $E$, and their difference is the latitude, the variation in wording that the entailment relation treats as irrelevant. The ledger gives the difference a name, and the leakage gives a test of whether the entailment clusters are adequate for a stated downstream use.

**Clustering and distances between partitions.** $\Lambda + \Delta$ is the distance between partitions of Rokhlin [48], used in clustering as the variation of information [38] (Proposition 3). It is the sum of the two directions of conditional entropy, and the ledger keeps the directions apart.

**Steganography and watermarking.** Theorem 8 is the arithmetic-coding construction of provably secure steganography [10, 30], used with language models by Ziegler et al. [67] and made efficient by minimum entropy coupling in Schroeder de Witt et al. [49]. Its capacity is the entropy of the cover given what must be preserved; the observation here is that "what must be preserved" can be the meaning, and that the capacity is then the latitude.

**Codon usage.** The dependence of synonymous codon choice on base composition [42, 54], on gene expression [26, 52] and its summary by an effective number of codons [61] or by Shannon entropy [66] are classical. The genetic code has been treated as a channel before [64]. We are not aware of an earlier split of the information a codon carries about its origin into levels of meaning by the chain rule, with the synonymous part identified as leakage; the numbers of Section 6.3 are new in that sense, and nothing in them is surprising to a molecular biologist.

## 8 Limits and open problems

**A partition is not all of meaning.** The ledger accounts for a chosen $W$ or a chosen $Y$. It does not say that either is what a speaker intends. Theorem 5(d) is a proof of this limitation, not a way around it: every meaning in the theory is meaning for something. Meaning as intention, as truth-conditions or as reference is outside the scope.

**Meaning-preserving is relative to the map.** The embedding of Section 6.4 preserves the amino-acid sequence, which is the meaning map of this paper. It does not preserve everything biology cares about, and synonymous changes can affect expression, folding and function [34, 47]. A purpose that includes these has a finer meaning and a smaller latitude (Theorem 4(d)), and the capacity of the channel is that smaller latitude. The theorem is exact for the cover law it is given; choosing a cover law that is acceptable to the host is outside it.

**Independence.** The coding theorems assume independent draws. Text is not independent, and neither are the codons of a gene. The ledger identities hold for any joint law, including a joint law over blocks or over stationary processes with entropy rates; the coding statements would need the corresponding theorems for that setting. The estimates in Section 6 treat genes (or documents) as the independent unit and codons (or tokens) within them as exchangeable draws from the class law, which is an approximation.

**What the genome measurements show.** The shares of relevance carried by synonymous choice describe six chosen genomes and two purposes; other panels would give other numbers. That identity sits in the latitude is consistent with the known dependence of synonymous codon choice on base composition [42, 54] and is expected. The plastid, with 85 genes, is a small sample. The held-out test supports the decomposition on these genomes; it does not make the shares universal.

**What the English measurements show.** The sign of the balance $E - C = \Lambda - \Delta$ is not a property of English as such. In SemCor it is positive for lemmas, becomes negative when inflected words are the forms, is near zero for adjectives, and in wordnet inventories has no fixed sign (Sections 6.5 and 6.6). The meaning is a WordNet synset, a fine-grained inventory on which annotators themselves disagree; the corpus is American English of the 1960s; the whole-corpus values have not converged; and only the well-attested core, which is a restriction of the source, is stable. We tested one corpus and one family of inventories. A second sense-tagged corpus, in particular of another genre or language, would be the natural next test and was not available to us.

**Leakage depends on the model.** Leakage is estimated from data through models of $P(y \mid x)$ and $P(y \mid w)$, and a poor model can overstate or understate it. On the genomes we computed it from tabulated codon laws with samples of $10^5$ codons or more, and checked it against held-out genes. A study of leakage in language would need a design that respects the dependence within documents, which we did not attempt.

**Open problems.** (1) An extension of Theorem 8 to sources with memory and to receivers who observe the text through a noisy channel; latitude would then compete with error

correction for the same freedom. (2) A network version of the ledger, in which several speakers share meanings and the doubt of one is the latitude of another. (3) A calibrated estimator of the leakage of a declared meaning under dependence, with a guarantee. (4) A test of the vocabulary law against languages with known lexicons, where the effective vocabulary $2^E$ can be compared with the size of the lexicon.

## 9 Conclusion

Meaning was set aside in 1948 because the engineering problem did not need it. The ledger takes it back with no new mathematics: $E = I + \Lambda$ and $C = I + \Delta$. What is said differs from what is meant by the freedom of the speaker minus the doubt of the listener. Latitude costs vocabulary and, in an idealized model, buys a channel; doubt costs questions. Which meaning one uses is a choice of purpose, and the leakage says how much that choice discards. On the genetic code the numbers are definite: about a quarter of the entropy of a codon is latitude, and in the genomes measured nearly nine tenths of what a codon says about where it came from is written there. In English the balance between latitude and doubt changes sign with the definition of the form, which is a reason to state that definition whenever the ledger is reported.

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## A Data, software and reproducibility

**Sequence data.** The coding sequences are example files distributed with public software and read without modification. The *E. coli* K-12 MG1655, *S. aureus* NCTC 8325, *S. argenteus* and *S. agnetis* files are the nucleotide FASTA examples of the CodonU package; the *C. simulans* PES1 record is the GenBank file distributed with the codoff package; the *A. thaliana* plastid record (NC_000932) is from the Biopython test suite [13]; the list of highly expressed *E. coli* genes is the example file of the freqgen package. SHA-256 sums of the files as used are in `data/CHECKSUMS.txt` of the archive. The accessions of Table 4 are those in the file headers.

**Corpus and inventories.** SemCor is read through NLTK [6] from the `semcor` and `wordnet` packages (WordNet 3.0). Documents whose path begins with `brown1` or `brown2` are used; `brownv` (verbs only) is not. The wordnets of other languages are the tab files of the `omw-1.4` package [9]; only lines of type `lemma` are used.

**Software.** Python with NumPy, SciPy, pandas and Biopython; LaTeX with pgfplots. The versions used are in `data/ENVIRONMENT.txt`. Random generators are seeded in each script; the seed of every experiment is in its script.

**Scripts.** All code is in the `code` folder of the archive; `README.md` gives the order in which to run it.

| Script | Purpose |
|---|---|
| `ledgerlib.py` | definitions of the ledger, estimators, certified interval |
| `fetch_data.py` | downloads the raw files and writes their SHA-256 sums |
| `dna_experiment.py` | ledger, depth and identity of the genomes (Section 6) |
| `dna_identification.py` | held-out test of the relevance ledger |
| `embedding_demo.py` | latitude channel on real genes (Theorem 8 and Proposition 9) |
| `extract_semcor.py` | reads SemCor through NLTK |
| `semcor_experiment.py` | SemCor ledger, sensitivity analysis, profiles (Section 6.5) |
| `semcor_subsample.py` | sample-size dependence of the plug-in ledger |
| `inventory_experiment.py` | type-level ledger of 32 wordnets (Section 6.6) |
| `verification.py` | randomized checks of the identities (Table 11) |
| `estimator_validation.py` | estimators on a law of known truth (Section 5) |
| `record_environment.py` | software versions |
| `make_assets.py` | writes every table body, plot data file and number macro of this document |

## B Notation

| Symbol | Meaning | Where |
|---|---|---|
| $W, X, Y$ | meaning, form, purpose (consequence) | Definitions 1 and 3 |
| $C = H(W)$ | content | Definition 2 |
| $E = H(X)$ | expression | Definition 2 |
| $\Lambda = H(X \mid W)$ | latitude | Definition 2 |
| $\Delta = H(W \mid X)$ | doubt | Definition 2 |
| $I = I(W;X)$ | conveyance | Definition 2 |
| $b(x)$, $M_Y = b(X)$ | belief about $Y$ given $x$; meaning of $X$ for purpose $Y$ | Definition 3 |
| $\mathcal{L}_Y(W) = I(X;Y \mid W)$ | leakage of the meaning $W$ about $Y$ | Definition 4 |
| $G$, $W_k$ | coarser meanings; levels of depth | Proposition 6 |
| $Q(x^n \mid w^n)$ | cover law of the latitude channel | Section 4.2 |
| $\mu, \sigma^2$ | mean and variance of $-\log Q(X^n \mid w^n)$ | Theorem 8 |
| $\tau$ | minimum count defining the well-attested core | Section 6.5 |
| $d(W,X) = \Lambda + \Delta$ | distance between meaning and form | Proposition 3 |

**Table 12.** The type-level ledger of 32 wordnets.

| Code | Language | Synsets | Pairs | $\Lambda$ | $\Delta$ | $\Lambda - \Delta$ |
|---|---|---|---|---|---|---|
| eng | English | 117,659 | 206,941 | 1.054 | 0.834 | 0.219 |
| fin | Finnish | 116,763 | 189,183 | 0.904 | 0.949 | -0.045 |
| tha | Thai | 73,350 | 95,517 | 0.528 | 0.336 | 0.192 |
| fra | French | 59,091 | 99,096 | 0.995 | 1.620 | -0.624 |
| jpn | Japanese | 57,184 | 158,048 | 2.380 | 1.147 | 1.233 |
| ron | Romanian | 56,026 | 84,636 | 0.796 | 1.225 | -0.428 |
| cat | Catalan | 45,826 | 70,579 | 0.844 | 0.948 | -0.104 |
| por | Portuguese | 43,895 | 67,585 | 0.844 | 0.828 | 0.015 |
| slv | Slovene | 42,583 | 70,929 | 1.038 | 1.424 | -0.386 |
| cmn | Chinese (Mandarin) | 42,312 | 79,809 | 1.188 | 0.606 | 0.582 |
| spa | Spanish | 38,512 | 57,639 | 0.788 | 1.139 | -0.351 |
| ind | Indonesian | 38,085 | 106,578 | 2.061 | 2.515 | -0.454 |
| zsm | Malay | 36,911 | 105,017 | 2.065 | 2.576 | -0.512 |
| ita | Italian | 35,001 | 63,075 | 1.168 | 1.007 | 0.161 |
| pol | Polish | 33,826 | 52,336 | 0.868 | 0.311 | 0.557 |
| nld | Dutch | 30,177 | 60,062 | 1.392 | 0.717 | 0.675 |
| eus | Basque | 29,413 | 48,931 | 0.967 | 1.369 | -0.403 |
| hrv | Croatian | 23,120 | 47,871 | 1.531 | 1.121 | 0.411 |
| glg | Galician | 19,312 | 27,133 | 0.676 | 0.356 | 0.320 |
| slk | Slovak | 18,507 | 43,994 | 1.604 | 0.912 | 0.691 |
| ell | Greek | 18,049 | 24,102 | 0.559 | 0.635 | -0.076 |
| ita-iwn | Italian (second wordnet) | 15,563 | 24,135 | 0.903 | 0.470 | 0.434 |
| arb† | Arabic | 9,916 | 37,342 | 2.843 | 1.815 | 1.028 |
| lit† | Lithuanian | 9,462 | 16,021 | 0.923 | 0.731 | 0.193 |
| swe† | Swedish | 6,796 | 6,904 | 0.032 | 0.335 | -0.303 |
| heb† | Hebrew | 5,448 | 6,872 | 0.461 | 0.803 | -0.342 |
| bul† | Bulgarian | 4,959 | 8,935 | 1.099 | 0.570 | 0.529 |
| isl† | Icelandic | 4,951 | 16,004 | 2.075 | 0.679 | 1.397 |
| als† | Albanian | 4,675 | 9,599 | 1.255 | 1.043 | 0.213 |
| dan† | Danish | 4,476 | 5,859 | 0.573 | 0.564 | 0.008 |
| nob† | Norwegian (Bokmål) | 4,455 | 5,586 | 0.482 | 0.596 | -0.114 |
| nno† | Norwegian (Nynorsk) | 3,671 | 4,762 | 0.550 | 0.697 | -0.147 |

Logarithms are to base 2 and information is in bits. $\mathbb{P}$ is probability and E expectation.

## C The ledger of the wordnets

Table 12 lists the type-level ledger of Section 6.6 for every wordnet of the Open Multilingual Wordnet package with at least 1,000 pairs, and for English WordNet 3.0. A dagger marks wordnets with fewer than 10,000 synsets, which are not in Figure 11. Each distinct (synset, lemma) pair is counted once; $\Lambda$ and $\Delta$ are in bits. The resources differ in how they were built, so the rows are comparable as descriptions of the resources and not as measurements of languages.